Monte Carlo Methods

Waffle Univ. Reinforcement Learning Seminar

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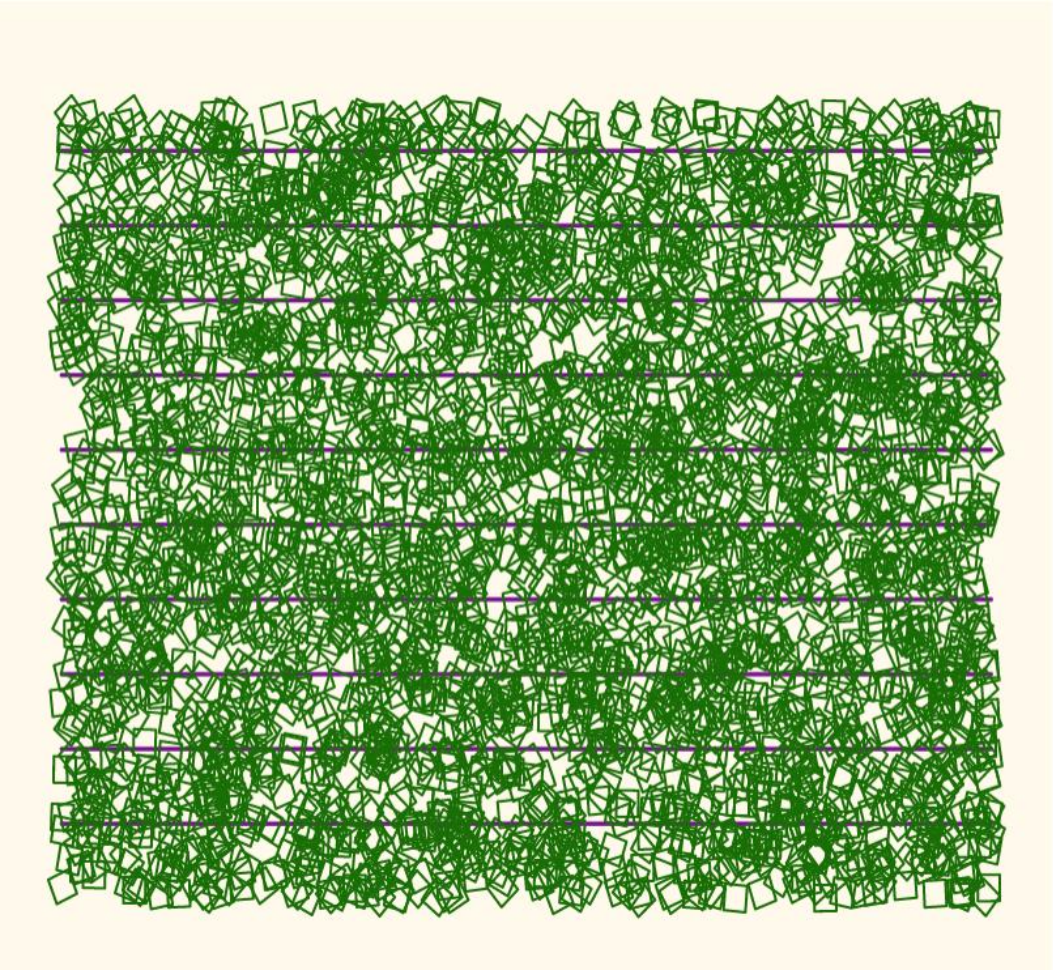
A long time ago..



뷔퐁의 바늘문제 변형 (Buffon's needle)

일정한 간격의 평행선 위에 바늘이 아닌
정사각형을 떨어뜨린다면 확률은 어떻게 될까?

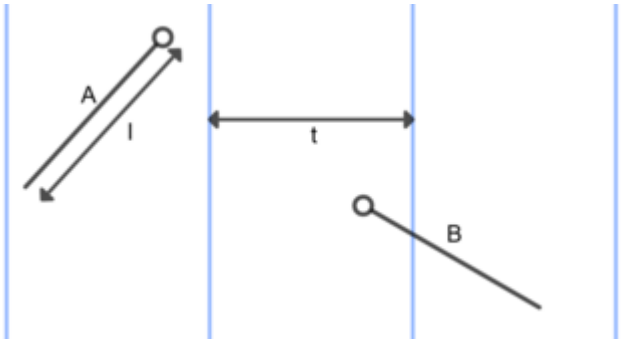
d : 평행선의 너비
l : 정사각형의 대각선 길이
$l = kd$

k = 0.5

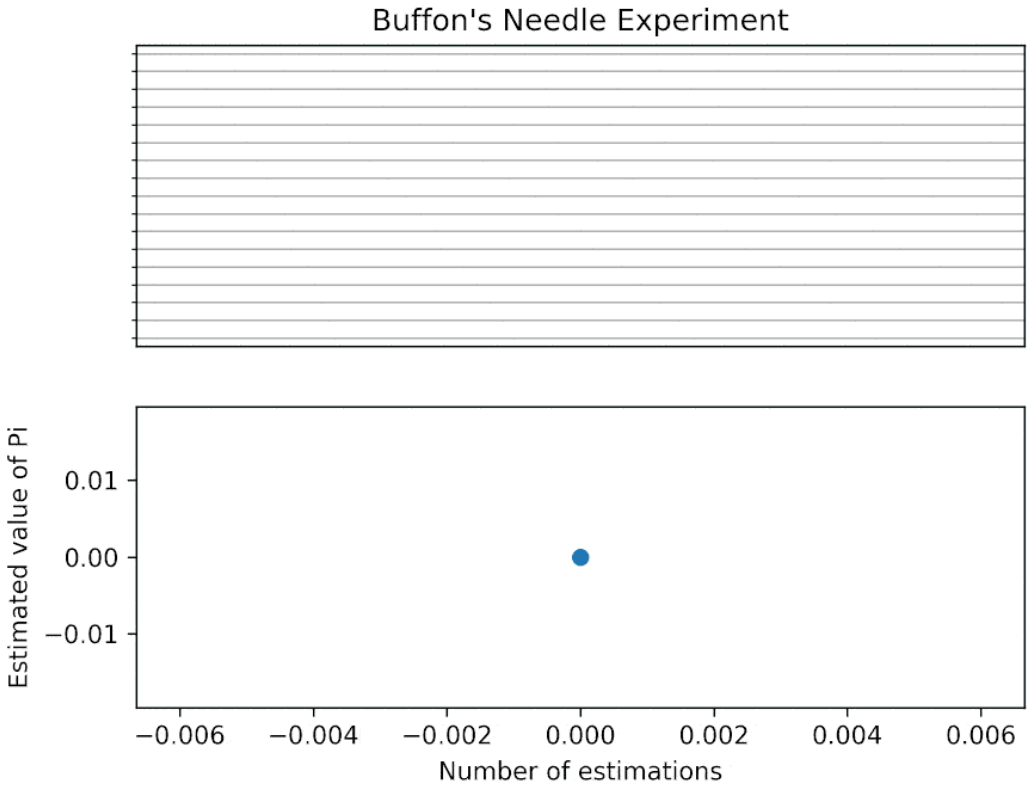
Drop

정사각형의 총 수(N)	3000
직선과 만나는 정사각형의 수(S)	1364
상대도수(S/N)	0.4547

Buffon's needle problem



$$p = \frac{2}{\pi} \cdot \frac{l}{t}.$$



$$\pi \approx \frac{2l \cdot n}{th}.$$

Monte Carlo method

According to the weak law of large numbers,

For the case where $X_1, X_2, \ldots$ is an infinite sequence of independent and identically distributed random variables with expected value

$$E(X_1) = E(X_2) = \cdots = \mu,$$

Then

$$\overline{X}_n = \frac{1}{n}(X_1 + \cdots + X_n)$$

Converges to the expected value:

$$\lim_{n\to\infty} \Pr\Big(|\overline{X}_n - \mu| < \varepsilon \Big) = 1.$$

Monte Carlo method

Simply, Monte Carlo method gives an estimate for μ by running n simulations and averaging the simulations' results.

Pros: we can gain μ for which we does not have a formula available to compute it.

Cons: It requires tremendous resources.
(i.e. computational cost, time, etc.)

Recap on Week2 Seminar

How to represent

Policies and Value Functions

	상태/행동 공간 크기가 작다	상태/행동 공간 크기가 크다
모델을 알고 있다	**Dynamic Programming**	**Approximation Method**
모델을 모른다	**Monte Carlo, TD SARSA, Q-Learning**	**DQN, Actor-Critic, PPO**

신경망으로 접근하기도 함

Knowledge on Model = Do we know the dynamics of the MDP?

$$p(s', r \mid s, a) \doteq \Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\},$$

Structure of the material

on-policy method

1. Monte Carlo Prediction
2. Monte Carlo Estimation of Action Values
3. Monte Carlo Control
4. Monte Carlo Control without Exploring Starts

off-policy method

5. Off-policy Prediction via Importance Sampling
6. Incremental Implementation
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9. Per-decision Importance Sampling

1. Monte Carlo Prediction

How can we learn the value of the state $v_\pi(s)$ for a given policy without the knowledge of model?

It's simple!

Just average the returns observed after visit to the state.

As more returns are observed, the average should converge to the expe value.

1. Monte Carlo Prediction

First-visit MC method vs Every-visit MC method

First-visit MC method

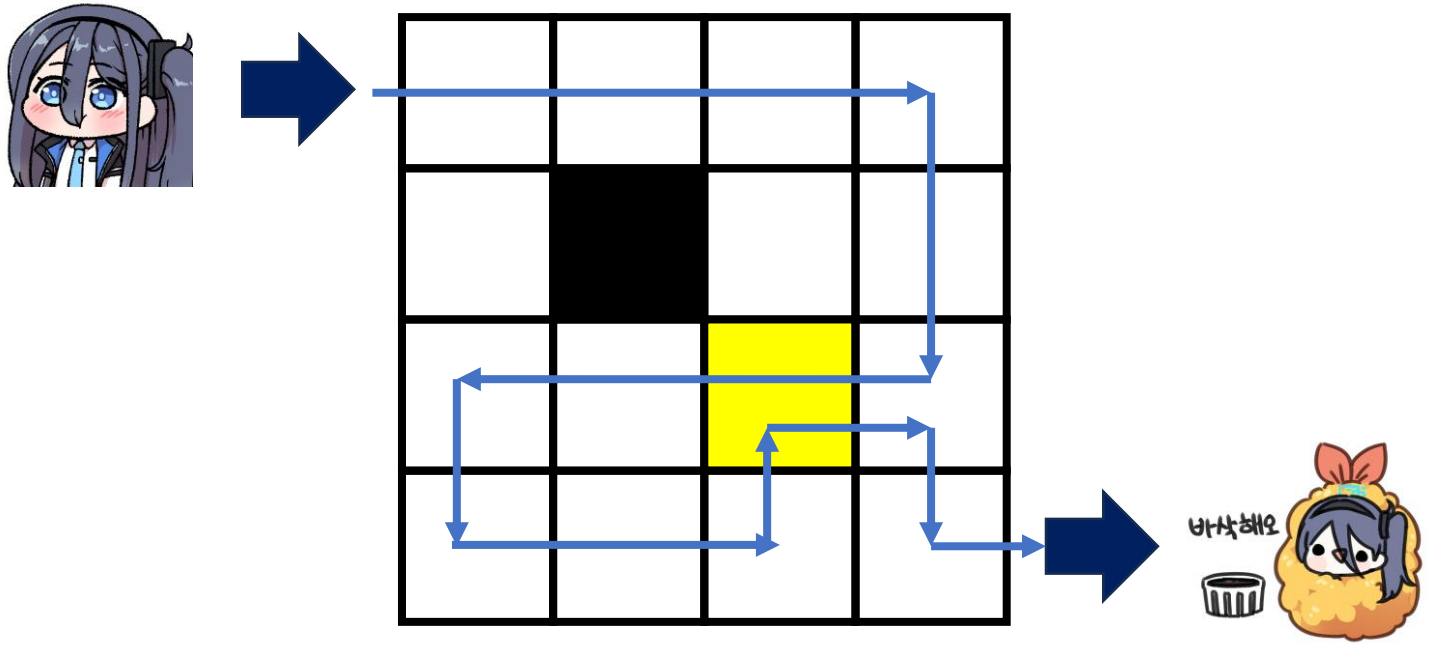
- Estimates $v_\pi(\mathrm{s})$ as the average of the returns following first visits to s

Every-visit MC method

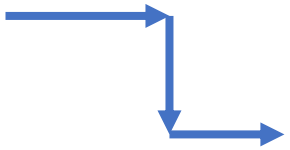
- Estimates $v_\pi(\mathrm{s})$ as the average of the returns following all visits to s

First-visit MC method vs Every-visit MC method

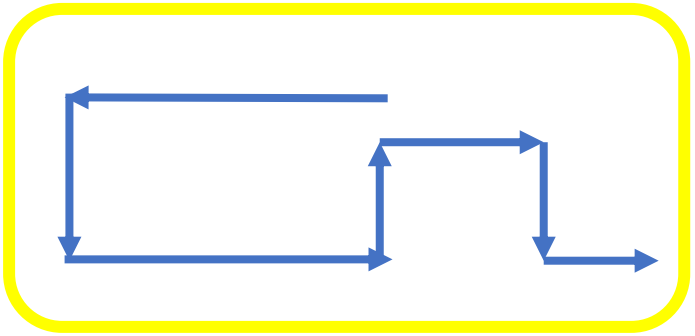
Simple grid world maze example



Every-visit MC



First-visit MC



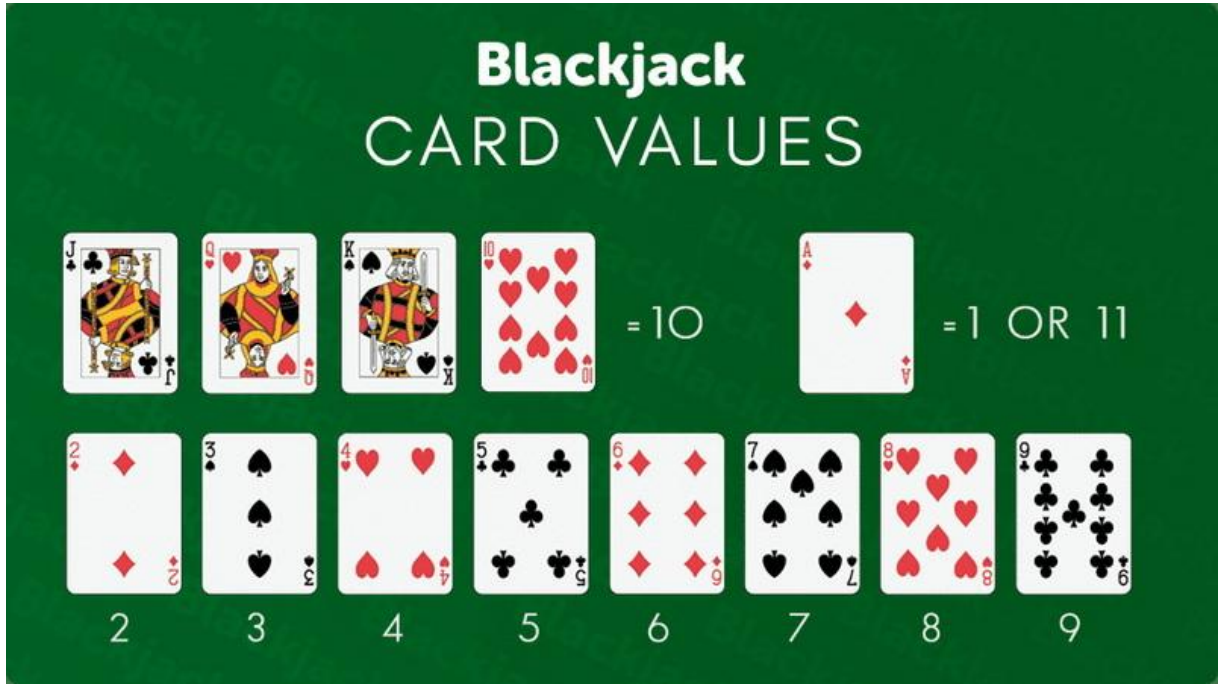
Algorithm for First-visit MC prediction

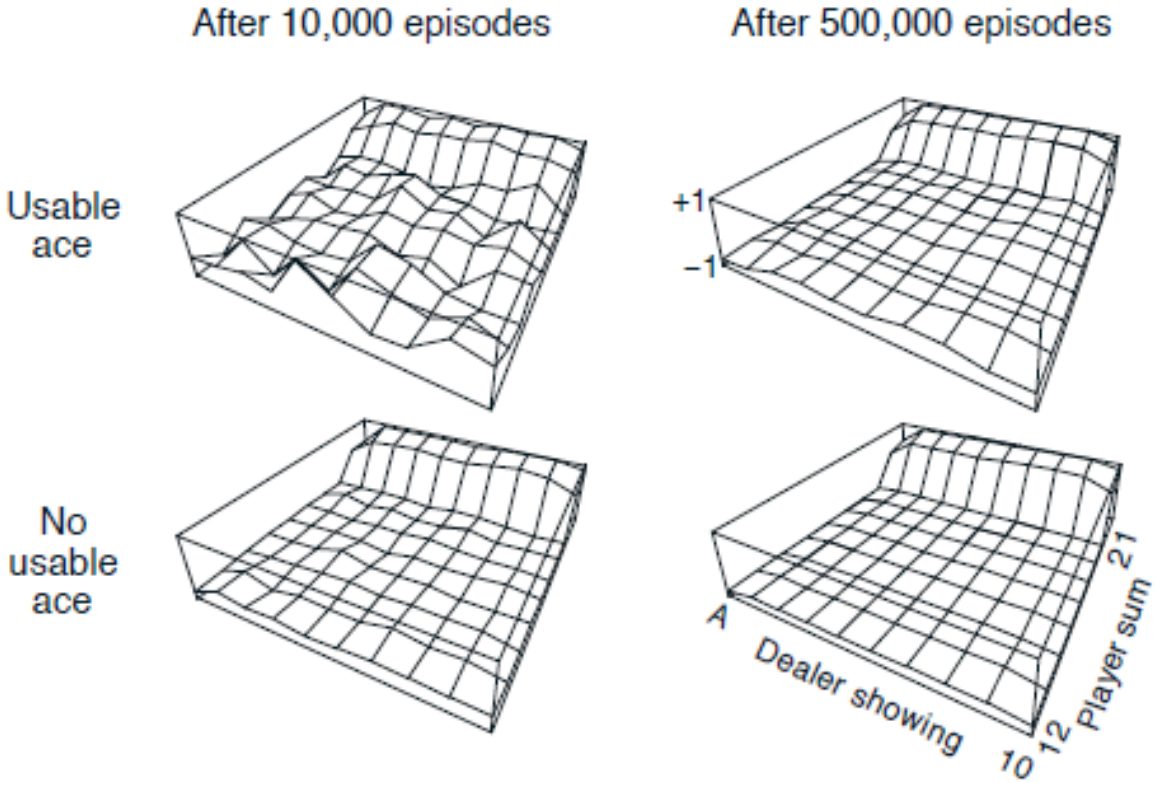
First-visit MC prediction, for estimating $V \approx v_\pi$

Input: a policy π to be evaluated
Initialize:
  $V(s) \in \mathbb{R}$, arbitrarily, for all $s \in \mathcal{S}$
  $Returns(s) \leftarrow$ an empty list, for all $s \in \mathcal{S}$

Loop forever (for each episode):
  Generate an episode following π: $S_0, A_0, R_1, S_1, A_1, R_2, \ldots, S_{T-1}, A_{T-1}, R_T$
  $G \leftarrow 0$
  Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
    $G \leftarrow \gamma G + R_{t+1}$
    Unless S_t appears in $S_0, S_1, \ldots, S_{t-1}$:
      Append G to $Returns(S_t)$
      $V(S_t) \leftarrow \text{average}(Returns(S_t))$

Blackjack Example





Although we know about the environment fully, it is not easy to determine dynamics.
It becomes easier with MC method. (computational expense of estimating the value of a single state is independent of the # of states)

2. Monte Carlo Estimation of Action Values

When a model is not available,

it is more useful to estimate action values, not state values.

Why?

With a model, state values alone are sufficient to determine policy.

But without a model, it's not. We need action value to determine policy.

But there is a problem that many state-action pairs may never be visited.

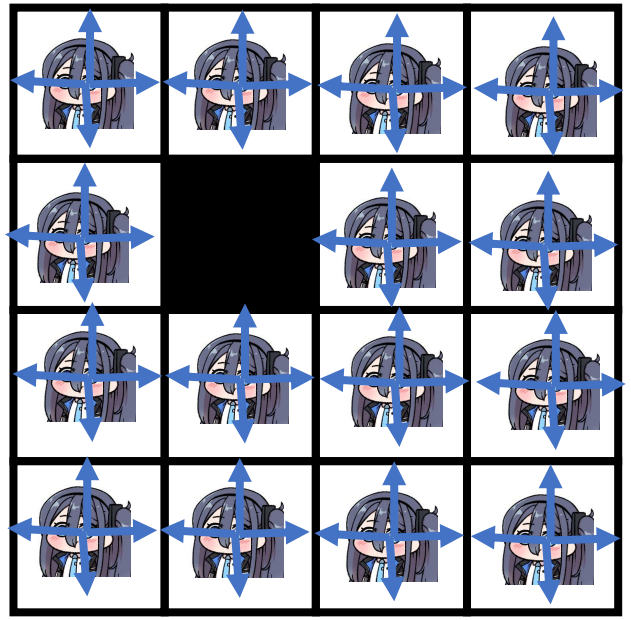
It will only evaluate value for visited state-action pairs while other actions will not improve with experience. => We need EXPLORATION!

2. Monte Carlo Estimation of Action Values

Exploring Starts:

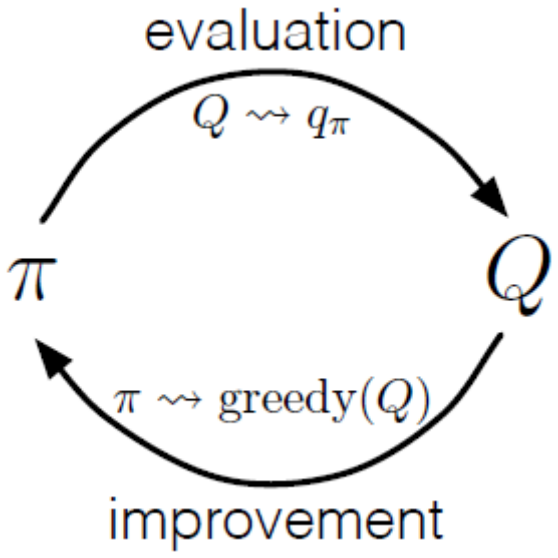
Episodes start in a state-action pair, and that every pair has a non-zero probability of being selected as the start.

Simple grid world maze example



3. Monte Carlo Control

Just like generalized policy iteration (GPI)



$$\pi_0 \xrightarrow{\text{E}} q_{\pi_0} \xrightarrow{\text{I}} \pi_1 \xrightarrow{\text{E}} q_{\pi_1} \xrightarrow{\text{I}} \pi_2 \xrightarrow{\text{E}} \cdots \xrightarrow{\text{I}} \pi_* \xrightarrow{\text{E}} q_*,$$

3. Monte Carlo Control

Policy improvement is done by making the policy greedy with respect to the current value function

$$\pi(s) \doteq \arg\max_a q(s,a).$$

$$\begin{aligned} q_{\pi_k}(s, \pi_{k+1}(s)) &= q_{\pi_k}(s, \arg\max_a q_{\pi_k}(s,a)) \\ &= \max_a q_{\pi_k}(s,a) \\ &\geq q_{\pi_k}(s, \pi_k(s)) \\ &\geq v_{\pi_k}(s). \end{aligned}$$

3. Monte Carlo Control

We've been assumed that policy evaluation operates on an infinite numbers of episodes, but it's not practical.

Solutions

1. Take steps until bounds on the magnitude and probability of error in the estimates are sufficiently small.

2. Give up trying to complete policy evaluation before returning to policy improvement. (e.g. value iteration)

Algorithm for Monte Carlo ES

Monte Carlo ES (Exploring Starts), for estimating $\pi \approx \pi_*$

Initialize:
- $\pi(s) \in \mathcal{A}(s)$ (arbitrarily), for all $s \in \mathcal{S}$
- $Q(s,a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$
- $Returns(s,a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Loop forever (for each episode):
- Choose $S_0 \in \mathcal{S}$, $A_0 \in \mathcal{A}(S_0)$ randomly such that all pairs have probability > 0
- Generate an episode from S_0, A_0, following π: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
- $G \leftarrow 0$
- Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
  - $G \leftarrow \gamma G + R_{t+1}$
  - Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1 \ldots, S_{t-1}, A_{t-1}$:
    - Append G to $Returns(S_t, A_t)$
    - $Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$
    - $\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$

4. Monte Carlo Control without Exploring Starts

However, assumption of ES is not realistic.

How can we avoid this?

We should make all actions are selected infinitely often.

There are two approaches.

on-policy methods & off-policy methods

In this chapter, we consider on-policy control methods.

4. Monte Carlo Control without Exploring Starts

Soft policy: $\boldsymbol{\pi}(\boldsymbol{a}|\boldsymbol{s}) > \mathbf{0}$ for all $s \in \boldsymbol{\mathcal{S}}$ and all $\boldsymbol{a} \in \boldsymbol{\mathcal{A}}(\boldsymbol{s})$

$\boldsymbol{\varepsilon}$-soft policy: $\boldsymbol{\pi}(\boldsymbol{a}|\boldsymbol{s}) \geq \frac{\boldsymbol{\varepsilon}}{|\boldsymbol{\mathcal{A}}(\boldsymbol{s})|}$ for all $s \in \boldsymbol{\mathcal{S}}$ and all $\boldsymbol{a} \in \boldsymbol{\mathcal{A}}(\boldsymbol{s})$

$\boldsymbol{\varepsilon}$-greedy policy

for greedy action: $\boldsymbol{\pi}(\boldsymbol{a}|\boldsymbol{s}) = 1 - \boldsymbol{\varepsilon} + \frac{\boldsymbol{\varepsilon}}{|\boldsymbol{\mathcal{A}}(\boldsymbol{s})|}$

for non-greedy action: $\boldsymbol{\pi}(\boldsymbol{a}|\boldsymbol{s}) = \frac{\boldsymbol{\varepsilon}}{|\boldsymbol{\mathcal{A}}(\boldsymbol{s})|}$

4. Monte Carlo Control without Exploring Starts

Without ES, we can't simply improve policy by making it greedy, because it would prevent future exploration of nongreedy actions.

But we can make policy to be moved toward greedy by using ε-greedy policy.

4. Monte Carlo Control without Exploring Starts

For any ε-soft policy π, any ε-greedy policy with respect to q_π is guaranteed to be better than or equal to π

Also, we can show that ε-greedy policy is optimal among ε-soft policies. (not shown here)

$$
\begin{aligned}
q_\pi(s, \pi'(s)) &= \sum_a \pi'(a|s) q_\pi(s,a) \\
&= \frac{\varepsilon}{|\mathcal{A}(s)|} \sum_a q_\pi(s,a) + (1-\varepsilon) \max_a q_\pi(s,a) \\
&\geq \frac{\varepsilon}{|\mathcal{A}(s)|} \sum_a q_\pi(s,a) + (1-\varepsilon) \sum_a \frac{\pi(a|s) - \frac{\varepsilon}{|\mathcal{A}(s)|}}{1-\varepsilon} q_\pi(s,a) \\
&= \frac{\varepsilon}{|\mathcal{A}(s)|} \sum_a q_\pi(s,a) - \frac{\varepsilon}{|\mathcal{A}(s)|} \sum_a q_\pi(s,a) + \sum_a \pi(a|s) q_\pi(s,a) \\
&= v_\pi(s).
\end{aligned}
$$

4. Monte Carlo Control without Exploring Starts

ε-greedy policy π is optimal among ε-soft policy

$$
\begin{aligned}
\tilde{v}_*(s) &= (1-\varepsilon)\max_a \tilde{q}_*(s,a) + \frac{\varepsilon}{|\mathcal{A}(s)|}\sum_a \tilde{q}_*(s,a) \\
&= (1-\varepsilon)\max_a \sum_{s',r} p(s',r\,|\,s,a)\Big[r+\gamma\tilde{v}_*(s')\Big] \\
&\quad + \frac{\varepsilon}{|\mathcal{A}(s)|}\sum_a\sum_{s',r} p(s',r\,|\,s,a)\Big[r+\gamma\tilde{v}_*(s')\Big].
\end{aligned}
$$

$$
\begin{aligned}
v_\pi(s) &= (1-\varepsilon)\max_a q_\pi(s,a) + \frac{\varepsilon}{|\mathcal{A}(s)|}\sum_a q_\pi(s,a) \\
&= (1-\varepsilon)\max_a \sum_{s',r} p(s',r\,|\,s,a)\Big[r+\gamma v_\pi(s')\Big] \\
&\quad + \frac{\varepsilon}{|\mathcal{A}(s)|}\sum_a\sum_{s',r} p(s',r\,|\,s,a)\Big[r+\gamma v_\pi(s')\Big].
\end{aligned}
$$

Algorithm of On-policy first-visit MC control for ε-soft policies

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$
Initialize:
  $\pi \leftarrow$ an arbitrary ε-soft policy
  $Q(s,a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$
  $Returns(s,a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Repeat forever (for each episode):
  Generate an episode following π: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
  $G \leftarrow 0$
  Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
    $G \leftarrow \gamma G + R_{t+1}$
    Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1 \ldots, S_{t-1}, A_{t-1}$:
      Append G to $Returns(S_t, A_t)$
      $Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$
      $A^* \leftarrow \text{argmax}_a Q(S_t, a)$ (with ties broken arbitrarily)
      For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \varepsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

5. Off-policy Prediction via Importance Sampling

Problem of On-policy method

All learning control methods face a dilemma

They seek to learn action values conditional on subsequent optimal behavior

but they need to behave non-optimally in order to explore all actions

The on-policy approach in the preceding section actually learns action values not for the optimal policy, but for a near-optimal policy that still explores.

5. Off-policy Prediction via Importance Sampling

Concept of Off-policy method

Target policy, π

The policy being learned and that becomes the optimal policy (student)

Behavior policy, b

The policy that is more exploratory and is used to generate behavior (teacher)

Using logged return of trajectory that follows behavior policy, estimate the action value of the target policy.

5. Off-policy Prediction via Importance Sampling

On-policy vs Off-policy

On-policy methods are

generally simpler and are considered first

Off policy methods are

require additional concepts and notation

often of greater variance and are slower to converge

more powerful and general

variety of additional uses in applications

5. Off-policy Prediction via Importance Sampling

Assumption of coverage

In order to use episodes from b to estimate values for π, we require that every action taken under π is also taken, at least occasionally, under b

Or just simply,
$\pi(a|s) > 0$ implies $b(a|s) > 0$.

5. Off-policy Prediction via Importance Sampling

Importance sampling

For given state-action trajectory occurring under any policy $\boldsymbol{\pi}$,

$$\begin{aligned}
&\Pr\{A_t, S_{t+1}, A_{t+1}, \ldots, S_T \mid S_t, A_{t:T-1} \sim \pi\} \\
&\quad = \pi(A_t|S_t)p(S_{t+1}|S_t, A_t)\pi(A_{t+1}|S_{t+1}) \cdots p(S_T|S_{T-1}, A_{T-1}) \\
&\quad = \prod_{k=t}^{T-1} \pi(A_k|S_k)p(S_{k+1}|S_k, A_k),
\end{aligned}$$

Importance sampling ratio

$$\rho_{t:T-1} \doteq \frac{\prod_{k=t}^{T-1} \pi(A_k|S_k)p(S_{k+1}|S_k, A_k)}{\prod_{k=t}^{T-1} b(A_k|S_k)p(S_{k+1}|S_k, A_k)} = \prod_{k=t}^{T-1} \frac{\pi(A_k|S_k)}{b(A_k|S_k)}.$$

5. Off-policy Prediction via Importance Sampling

Importance sampling

The returns G_t what we have is the return of the behavior policy

From this, we can estimate the value of the target policy using importance-sampling ratio.

$$\mathbb{E}[\boldsymbol{\rho}_{t:T-1} G_t \mid S_t = \boldsymbol{s}] = \boldsymbol{v}_{\boldsymbol{\pi}}(\boldsymbol{s})$$

5. Off-policy Prediction via Importance Sampling

Ordinary importance sampling vs Weighted importance sampling

Ordinary importance sampling

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|}.$$

Weighted importance sampling

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1}}$$

variables

$\mathcal{T}(s)$: set of all time steps in which state s is visited

$T(t)$: the first time of termination following time t

$\{G_t\}_{t \in \mathcal{T}(s)}$: returns that pertain to state s

$\{\rho_{t:T(t)-1}\}_{t \in \mathcal{T}(s)}$: corresponding importance-sampling ratios

Ordinary importance sampling vs Weighted importance sampling
- In perspective of bias & variance

Ordinary importance sampling is unbiased
Weighted importance sampling is biased

the variance of ordinary importance sampling is in general unbounded whereas in the weighted estimator the largest weight on any single return is one.

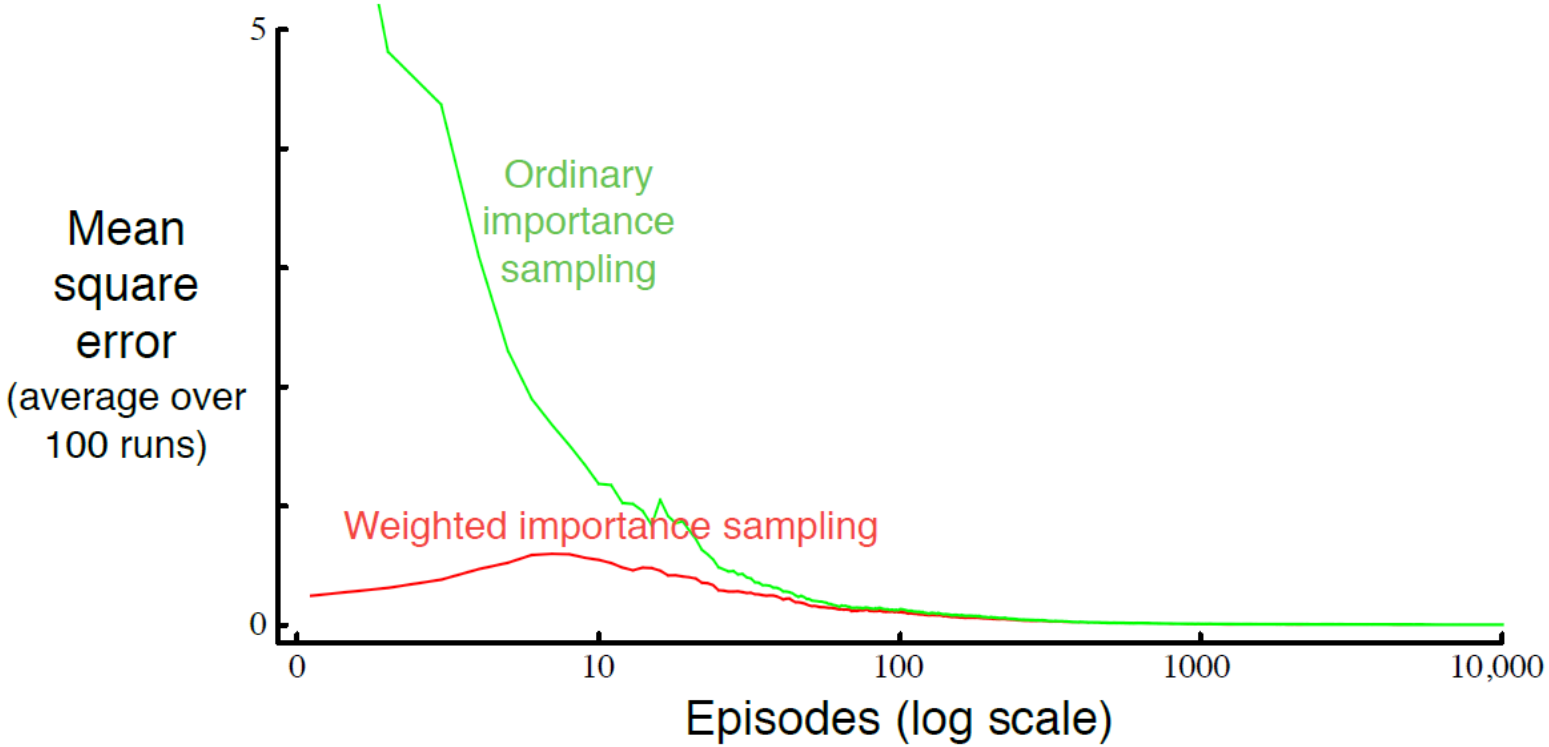
In fact, assuming bounded returns, the variance of the weighted importance-sampling estimator converges to zero even if the variance of the ratios themselves is infinite

In practice, the weighted estimator usually has dramatically lower variance and is strongly preferred

5. Off-policy Prediction via Importance Sampling

Example from book – example 5.4

evaluated the state in which the dealer is showing a deuce, the sum of the player's cards is 13, and the player has a usable ace

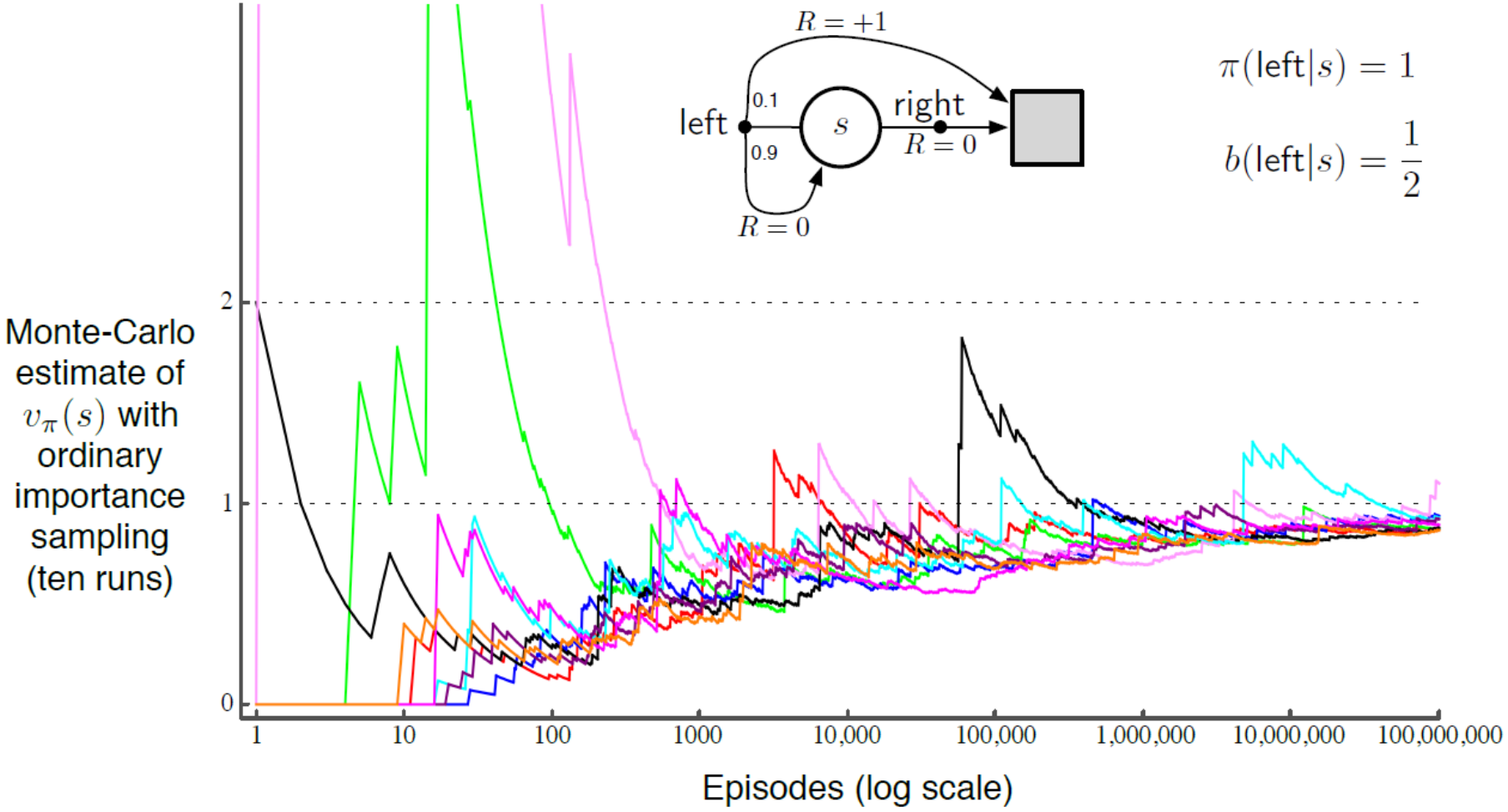


Behavior policy: 50/50 Hit or stick

Target policy: sticks when only the sum of the cards is 20 or 21

5. Off-policy Prediction via Importance Sampling

Example from book – example 5.5



5. Off-policy Prediction via Importance Sampling

Example from book – example 5.5

The variance of any random variable X is

$$\mathrm{Var}[X] \doteq \mathbb{E}\left[\left(X - \bar{X}\right)^2\right] = \mathbb{E}\left[X^2 - 2X\bar{X} + \bar{X}^2\right] = \mathbb{E}\left[X^2\right] - \bar{X}^2.$$

Thus, if the mean is finite, as it is in our case, the variance is infinite if and only if

$$\mathbb{E}_b\left[\left(\prod_{t=0}^{T-1} \frac{\pi(A_t|S_t)}{b(A_t|S_t)} G_0\right)^2\right]$$

Incremental Implementation

$$NewEstimate \leftarrow OldEstimate + StepSize\left[Target - OldEstimate\right].$$

to form a weighted average of the returns, a slightly different incremental algorithm is required

$$W_i = \rho_{t_i:T(t_i)-1}$$

$$V_n \doteq \frac{\sum_{k=1}^{n-1} W_k G_k}{\sum_{k=1}^{n-1} W_k}, \qquad n \geq 2$$

$$V_{n+1} \doteq V_n + \frac{W_n}{C_n}\left[G_n - V_n\right], \qquad n \geq 1,$$

$$C_{n+1} \doteq C_n + W_{n+1},$$

7. Off-policy Monte Carlo Control

Brief explanation of off-policy MC control

In off policy methods these two functions are separated

Advantage: the target policy may be deterministic (e.g., greedy) while the behavior policy can continue to sample all possible actions.

Requirements:

1. Assumption of coverage
2. Behavior policy is soft

Algorithm for off-policy MC

Off-policy MC control, for estimating $\pi \approx \pi_*$

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
- $Q(s,a) \in \mathbb{R}$ (arbitrarily)
- $C(s,a) \leftarrow 0$
- $\pi(s) \leftarrow \arg\max_a Q(s,a)$ (with ties broken consistently)

Loop forever (for each episode):
- $b \leftarrow$ any soft policy
- Generate an episode using b: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
- $G \leftarrow 0$
- $W \leftarrow 1$
- Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
  - $G \leftarrow \gamma G + R_{t+1}$
  - $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
  - $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)}[G - Q(S_t, A_t)]$
  - $\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken consistently)
  - If $A_t \neq \pi(S_t)$ then exit inner Loop (proceed to next episode)
  - $W \leftarrow W \frac{1}{b(A_t|S_t)}$

Potential problem:

If nongreedy actions are common, then learning will be slow

8. Discounting-aware Importance Sampling

Flat partial returns

Let us now consider an idea for avoiding this large extraneous variance

$$\bar{G}_{t:h} \doteq R_{t+1} + R_{t+2} + \cdots + R_h, \qquad 0 \leq t < h \leq T,$$

$$\begin{aligned}
G_t &\doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots + \gamma^{T-t-1} R_T \\
&= (1-\gamma) R_{t+1} \\
&\quad + (1-\gamma)\gamma \left(R_{t+1} + R_{t+2}\right) \\
&\quad + (1-\gamma)\gamma^2 \left(R_{t+1} + R_{t+2} + R_{t+3}\right) \\
&\quad \vdots \\
&\quad + (1-\gamma)\gamma^{T-t-2} \left(R_{t+1} + R_{t+2} + \cdots + R_{T-1}\right) \\
&\quad + \gamma^{T-t-1} \left(R_{t+1} + R_{t+2} + \cdots + R_T\right) \\
&= (1-\gamma) \sum_{h=t+1}^{T-1} \gamma^{h-t-1} \bar{G}_{t:h} \quad + \quad \gamma^{T-t-1} \bar{G}_{t:T}.
\end{aligned}$$

8. Discounting-aware Importance Sampling

Discounting-aware IS

For ordinary importance-sampling estimator

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \left((1-\gamma) \sum_{h=t+1}^{T(t)-1} \gamma^{h-t-1} \rho_{t:h-1} \bar{G}_{t:h} \;+\; \gamma^{T(t)-t-1} \rho_{t:T(t)-1} \bar{G}_{t:T(t)} \right)}{|\mathcal{T}(s)|}$$

For weighted importance-sampling estimator

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \left((1-\gamma) \sum_{h=t+1}^{T(t)-1} \gamma^{h-t-1} \rho_{t:h-1} \bar{G}_{t:h} \;+\; \gamma^{T(t)-t-1} \rho_{t:T(t)-1} \bar{G}_{t:T(t)} \right)}{\sum_{t \in \mathcal{T}(s)} \left((1-\gamma) \sum_{h=t+1}^{T(t)-1} \gamma^{h-t-1} \rho_{t:h-1} \;+\; \gamma^{T(t)-t-1} \rho_{t:T(t)-1} \right)}$$

9. Per-decision Importance Sampling

Per-decision IS

$$\begin{aligned}\rho_{t:T-1}G_t &= \rho_{t:T-1}\left(R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-t-1}R_T\right) \\ &= \rho_{t:T-1}R_{t+1} + \gamma\rho_{t:T-1}R_{t+2} + \cdots + \gamma^{T-t-1}\rho_{t:T-1}R_T\end{aligned}$$

$$\mathbb{E}\left[\frac{\pi(A_k|S_k)}{b(A_k|S_k)}\right] \doteq \sum_a b(a|S_k)\frac{\pi(a|S_k)}{b(a|S_k)} = \sum_a \pi(a|S_k) = 1.$$

$$\mathbb{E}[\rho_{t:T-1}R_{t+1}] = \mathbb{E}[\rho_{t:t}R_{t+1}]\,.$$

$$\mathbb{E}[\rho_{t:T-1}R_{t+k}] = \mathbb{E}[\rho_{t:t+k-1}R_{t+k}]\,.$$

$$\mathbb{E}[\rho_{t:T-1}G_t] = \mathbb{E}\left[\tilde{G}_t\right]$$

$$\tilde{G}_t = \rho_{t:t}R_{t+1} + \gamma\rho_{t:t+1}R_{t+2} + \gamma^2\rho_{t:t+2}R_{t+3} + \cdots + \gamma^{T-t-1}\rho_{t:T-1}R_T\,.$$

$$V(s) \doteq \frac{\sum_{t\in\mathcal{T}(s)}\tilde{G}_t}{|\mathcal{T}(s)|}$$

Thank You and Any Question?